

Activity Prediction Method for Nursing Care Records with Missing Entries

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Abstract

With the proliferation of Information and Communication Technology (ICT), recording systems for care activities using smartphones and tablets are becoming widespread in the field of nursing care. As a result, it is becoming possible to predict future caregiving activities using the care record history recorded in the systems. The prediction of future caregiving activity enables us to develop various caregiving applications, such as support for the preparation of future caregiving activity, detection of missing entries in nursing care records, and prior information for real-time caregiving activity recognition using sensors.

However, while the recording has become easier with the introduction of the recording systems, there are still many missing nursing care entries. When data with many missing entries is used as training data for activity prediction methods, the performance of the prediction methods will deteriorate.

In this paper, we propose a caregiving activity prediction method that is robust against missing entries. The proposed model has a module for correcting missing entries, and its intermediate output is used to estimate whether or not a certain caregiving activity will occur from the current time to one hour later using the caregiving activity records of the past T hours. We evaluated the effectiveness of the proposed method using data obtained from actual nursing care facilities.

1 Introduction

In recent years, the population in developed countries has been aging resulting in high demand for caregivers. In Japan, it is estimated that

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there will be only 1.3 persons in the workforce for every elderly person aged 65 or older by 2065 [4]. Besides, approximately 690,000 new caregivers are estimated to be needed in 2040 compared to 2019. Therefore, it is necessary to improve the efficiency of care work, and the digitization of nursing care records is attracting attention. In recent years, ICT has been introduced in nursing care facilities. An increasing number of nursing homes are using recording systems [1, 2, 3, 11] that allow care records to be entered using smartphones and tablets. The system allows each caregiver to record the care activities performed by the caregiver, such as eating and toileting support, in addition to recording measurement data such as the care receiver's body temperature and blood pressure. Figure 1 shows an example of binarized hourly care activity records, where "1" shows the time when the activity occurred and 0 shows the time when it did not occur. Using the recording system instead of paper records can reduce work time, and the recording system makes it easy to analyze the recorded data. For example, it is possible to analyze a care recipient's short- and long-term health status based on the data entered by the caregiver. In addition, it is possible to predict caregiving activities that will occur in the future by using the data stored in the recording system.

activity class	0:00~ 1:00	1:00~ 2:00	...	12:00~ 13:00	...
meal	0	0	...	1	...
excretion	1	0	...	0	...
⋮	⋮	⋮	...	⋮	...
Bathing	0	0	...	0	...

care records

Figure 1: An example of binarized hourly care activity records. "1" shows an entry of a care activity occurring at the corresponding time slot.

Many studies use caregiving records to predict future caregiving activity based on past caregiving activity [8, 13, 15]. If future caregiving activities can be predicted, caregivers can prepare for necessary care and work more efficiently. Furthermore, the prediction of caregiving activity can be used as an assist in the recording of caregiving activities. Although recording tasks have been made easier with the development of recording systems, caregivers face a large burden of recording tasks because they must perform multiple care tasks at the same time. The prediction of caregiving activity can be used to help simplify the input of the recording system. For example, when meal care is predicted to occur within an hour, the recording system can help input the corresponding entry to reduce the time required for input. In addition, if the predicted care activity is not entered, the system notifies the caregiver whether or not they forgot to record the activity, thus preventing missing recordings. In addition, the prediction of caregiving activity can be used for caregiving activity recognition. That is, prediction results using past caregiving ac-

tivities can be used as prior knowledge in fine-grained caregiving activity recognition using sensor data of caregivers. As above, the information in caregiving records is useful for predicting caregiving activity. However, because caregivers record while performing multiple tasks, it is difficult to record without missing entries. As a result of our analysis of the nursing care records, many missing entries in the nursing care records were available. When records with many missing entries are used to train an activity prediction method, its prediction performance deteriorates.

In this paper, we propose a method for predicting future caregiving activity using nursing care records including missing entries. We estimate whether or not a certain caregiving activity will occur from the current time to one hour later using the caregiving activity records for the past T hours. To deal with the problem of missing entries, the proposed method is designed to have a module for correcting missing entries and leverage the intermediate outputs of the module for the prediction. In the proposed method, nursing care records used for training are divided into two, one with a high probability of missing entries and the other with a low probability of missing entries. For the records with a low possibility of missing entries, we artificially create missing entries by discarding some entries. After that, the correction module trains to correct the discarded entries. The contributions of this work are as follows:

- We proposed a robust activity prediction method based on caregiving records with missing entries as training data.
- We created artificial missing entries and designed a module that trains to correct the created missing entries.
- We evaluated the effectiveness of the proposed method using data obtained from actual nursing care facilities.

2 Related Work

2.1 Nursing Care Recording Systems

Many care recording systems using smartphones and tablets have been developed to improve the efficiency of care recording tasks. Care Palette [1], developed by ND Software Co., Ltd, can record data by voice input and automatically record measurements in wireless connection with vital devices such as thermometers. Suguroku Tablet [3], developed by Wiseman Co., Ltd, can record nursing care activities such as eating and toileting, and the recorded information is shared with caregivers in real time. Fonlog [2, 11], developed by the Kyushu Institute of Technology, uses sensor data to recognize caregivers' activities and automatically suggests caregiving activities, such as eating and toileting, as candidates for input into caregiving records. In this paper, we use the caregiving activity records collected by Fonlog to predict caregiving activity.

2.2 Prediction of Caregiver and Care Receiver activity

If future important caregiving activities are predicted, caregivers can prepare for them. Therefore, many studies on predicting caregiver and care recipient activity have been actively studied. Uchimura et al. [15] proposed a model for predicting urination time and the amount of urine in the bladder based on the amount of water intake using nursing care records. Prashant [6] developed a wearable device that measures the amount of urine in the bladder using NIR light.

There are studies on the prediction of caregiving activity using actual caregiving records collected at nursing care facilities. Saitoh et al. [13] focused on the error between the actual time of excretion care and the time on the care record and proposed a method to predict the occurrence of excretion care using machine learning such as random forests and support vector machines. Matsuki et al. [12] analyzed caregiver records and sensors to predict the amount of activity during the day based on the caregiver's sleep status.

Inoue et al. [8] proposed a method to automatically generate caregiving records by recognizing caregivers' activities using sensor data from smartphones and to predict the activities that caregivers will take on the next day and the time of their activities using the generated caregiving records.

3 Analysis of Nursing Care Records

To design the prediction method, we analyze the periodicity of caregiving activity, the relationship between different caregiving activities, and the tendency of missing entries in actual caregiving records.

3.1 Overview of Care Record

We use nursing care records collected using "Fonlog" by 23 caregivers from May 1, 2018, to July 1, 2018, for 34 subjects at a nursing home in Fukuoka, Japan.

There are two types of records regarding caregiving activity collected by Fonlog: "Activities" and "Records". Table 1 shows the relationship between activities and records. For example, if a caregiver provides urination care to a recipient, the caregiver enters "Excretion" as an activity and enters values for the amount of urination, method of excretion, and so on as records. In addition, the caregiver's name (ID) is entered. In this paper, we analyze the "Excretion" activity, which has a significant impact on caregivers' care planning. We assume that the excretion activity is categorized into two activities: urination and defecation. The records of "Amount of urination" and "Amount of defecation" were used for the categorization. Among the "Excretion" activities, we assumed the one with the "Amount of urination" record as a new "Urination" activity and the one with the "Amount of defecation" record as a new "Defecation" activity.

Table 1: the relationship between activities and records

Activities	Records
Meal and medicine	Assistance with meals Amount of food (staple food) Amount of food (side dishes) Amount of water Medicine
Excretion	Assistance with excretion Excretion method Amount of defecation Amount of urination Condition of defecation
Nighttime user care	Contents Type
Daytime user care	Contents Type
Morning exercise	Radio calisthenics Morning exercise
Bathing	Assistance with bathing Bathing method
Outdoor activity	Outing location Type
Treatment	Contents Treatment area
Oral care	Intraoral cleaning Dentures cleaning
Family / guest response	Purpose
Change dressing assistance	Change dressing assistance

In this paper, an hourly occurrence of an activity is represented as a binary value, with 1 meaning the time when the activity occurred and 0 meaning the time when it did not occur. This binarized time series data is referred to as a binary occurrence time series.

3.2 Analysis of Periodicity of Care Records

Non-periodic caregiving activities such as “Defecation” are difficult for caregivers to prepare in advance. Therefore, caregivers can benefit a lot from predicting non-periodic caregiving activities. To evaluate the periodicity of each activity, we calculated the autocorrelation coefficient using the binary occurrence time series for each activity. Figure 2 shows the average of the autocorrelation coefficients over the 34 subjects. Figure 2 suggests that the “Meal and medicine” and “Daytime user care” activities show strong autocorrelations when the lag is 24, 48, or 72 hours. In contrast, the “Defecation” and “Urination” activities have few large

peaks, indicating their small autocorrelations. As a result, the “Meal and medicine,” “Nighttime user care”, and “Daytime user care” activities are regarded as periodic, while the “Defecation” and “Urination” activities are non-periodic activities.

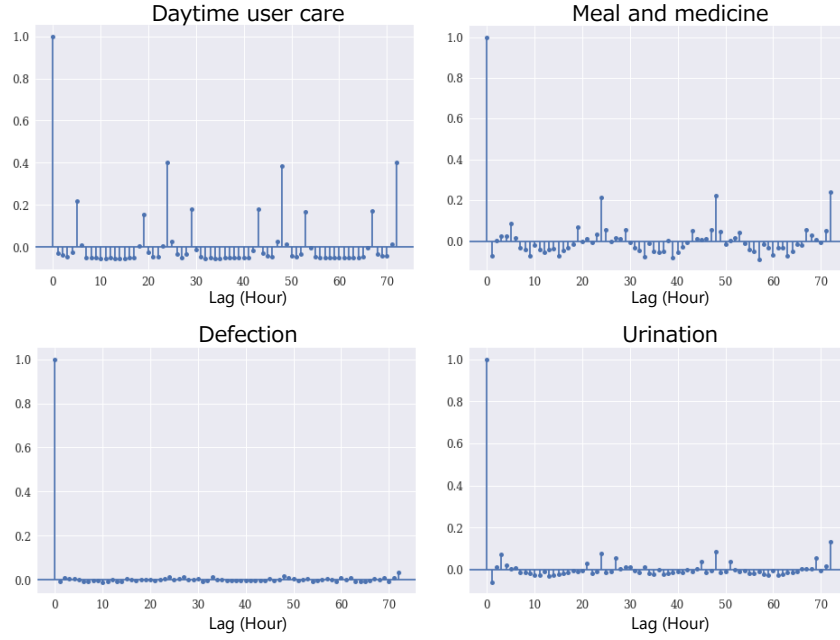


Figure 2: The average of the autocorrelation coefficients over 34 subjects for four activities

3.3 Analysis of the Relationship Between Different Caregiving Activities

To analyze the temporal relationship between different caregiving activities, we calculated cross-correlation coefficients using pairs of binary occurrence time series of different activities. Figure 3 shows the average of the cross-correlation coefficients over the 34 subjects. The “Defecation” and “Urination” activities have a strong cross-correlation when the lag is 0. This indicates that “Defecation” and “Urination” activities often occur at the same time. The “Meals and medication” activity and the “Daytime user care” activity have a strong cross-correlation when the lag is -4 and 2 hours. This suggests that the “Daytime user care” activity often occurs four hours before and two hours after the “Meal and medication” activity occurs. Furthermore, Figure 3 shows that, even in activities such as “Defecation” and “Urination”, which differ in timing depending on individuals, the temporal correlation with other activities seems to be available. These results indicate that the history of other activities is useful in predicting a certain activity.

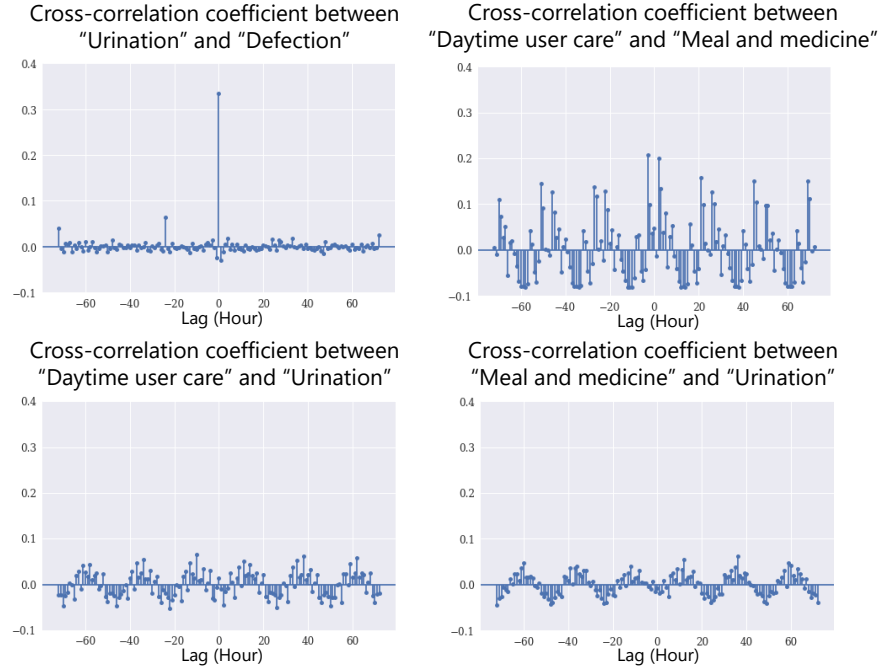


Figure 3: The average of the cross-correlation coefficients over 34 subjects of four activities.

3.4 Analysis of Missing Care Record Entries

It is difficult for caregivers to record care records without missing entries. Here, we analyze care records with missing entries for designing the proposed method. First, we counted the number of times each activity was performed within a 72-hour time window for 34 caregivers. Figure 4 shows the number of times an activity occurred within a 72-hour time window for 34 subjects. We can see that many 72-hour time windows have the number of occurrences much fewer than the average occurrences. This is considered that these time windows have missing entries.

Then we analyze the time periods with a low probability of missing entries. For the "Meals and medication" activity, when the number of occurrences within 72 hours is 9 or more (3 times per day), the probability of missing entries is considered low. Using this threshold, approximately 55% of the total time windows were identified as having a low probability of missing entries. Additionally, for the activities of "Urination" and "Defecation", when the occurrences of "Urination" within 72 hours are 15 times or more, and the occurrences of "Defecation" within 72 hours are 2 times or more, the probability of missing entries is considered low [9, 14]. In this case, the time windows with a low probability of missing entries for the "Urination" and "Defecation" activities accounted for approximately 17% and 45% of the total time windows, respectively.

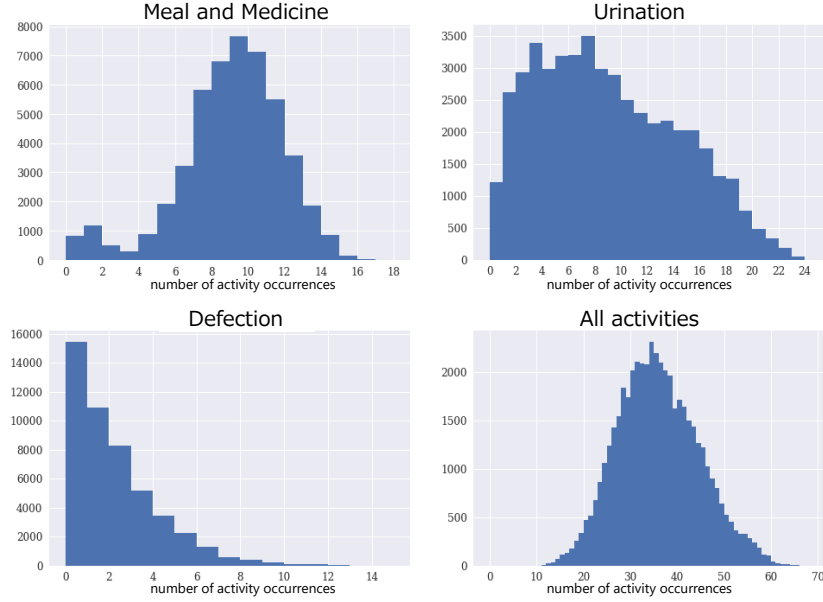


Figure 4: The distribution of the number of times an activity occurred within a 72-hour time window for 34 subjects

Assuming that missing entries occur at a similar rate for all the activities, the average proportion of time windows with a low probability for missing entries for these activities is considered approximately 39%, which is the average proportion over the three activities. Because, when the total number of occurrences of all activities within 72 hours is 38 or more, the proportion of time windows with the number of activity occurrences more than the threshold for all the windows is approximately 39%. Therefore, time windows with a total occurrence of 38 or more activities within 72 hours are considered periods with a low probability of missing entries.

4 Proposed Method

4.1 Overview

Figure 5 shows an overview of the proposed prediction model. The proposed model is designed to have a correction module that corrects missing entries and uses the intermediate outputs to predict the activity of care recipients. The proposed model mainly consists of the correction module and prediction module. The prediction module predicts whether or not the target activity will occur from the current time to one hour later. The correction module corrects missing entries in binary occurrence time series of the target activity of interest and outputs the corrected binary occurrence time series.

Initially, the correction module is trained. Afterward, its parameters are frozen, and then the prediction module is trained.

4.2 Prediction Model

The input to the prediction model is a binary time series of 72 hours in length. The output of the correction module is a binary time series of 72 hours in length. The output of the prediction module is a binary value indicating whether a certain activity will occur within one hour after the 72 hours used as input for the model.

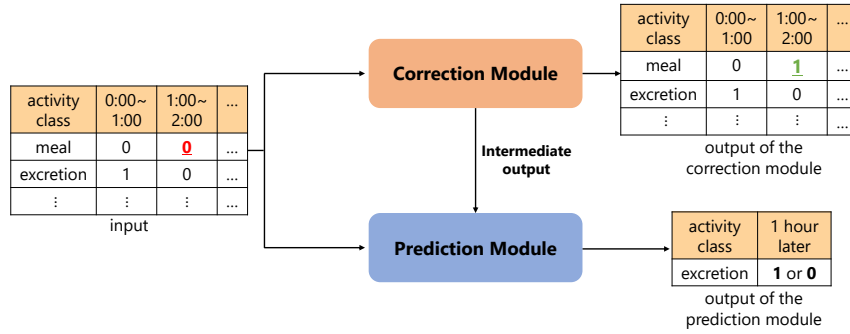


Figure 5: Overview of the proposed prediction model

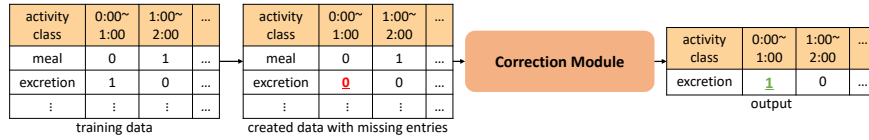


Figure 6: Overview of the training of the correction module

4.2.1 Correction Module

Figure 6 shows an overview of the training of the correction module. For training, the correction module uses only binary occurrence time series with a low probability of missing entries. We create records with missing entries by randomly invalidating activity occurrences with a certain probability. In other words, the value of 1 in the binary occurrence time series is changed to 0 with a certain probability. This method is described in detail in Section 4.3. The created binary occurrence time series containing the missing entries is fed into the correction module, which is trained to reconstruct the original data.

Figure 7 shows the architecture of the correction module. In general, the number of values of 1 representing the occurrence of an activity in a binary occurrence time series is smaller than the number of values of 0 representing the non-occurrence of the activity. To deal with the problem

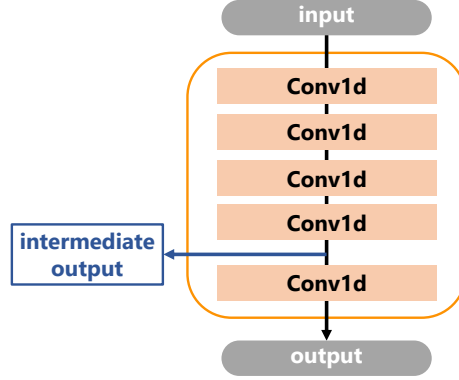


Figure 7: The architecture of the correction module

of the imbalanced data, we calculate the training loss by weighting the loss per class using the inverse of the number of data points belonging to the class [5, 7, 10, 16], enabling to train the model by emphasizing the value of 1, which represents the occurrence of activities.

Furthermore, we train the model so that the number of predicted activity occurrences will be similar to the actual number. Therefore, we calculate the ground-truth distribution of the number of occurrences of an activity of interest within 72 hours in advance. Specifically, the mean and standard deviation of the number of times the activity occurs every 72 hours in the data with a low probability of entries are calculated in advance to derive the cumulative density $F(x)$ for the estimate. By including the cumulative density into the loss function, the number of activity occurrences in the output data will be similar to the actual number of activity occurrences. The loss function $\mathcal{L}_{\mathcal{M}}$ of the correction module is expressed by the following equation.

$$\mathcal{L}_{\mathcal{M}} = \sum_{j=1}^B \sum_{k=0}^{71} f(y_{j,\tau-k}, j, \tau-k)$$

$$f(c, j, \tau-k) = \frac{1}{n_c} \{ (p_{j,\tau-k} - c)^2 + \beta \cdot F(\sum_{u=0}^{71} p_{j,\tau-u}) \}$$

where B is the batch size, $\mathbf{P}_j = [p_{j,\tau-71}, \dots, p_{j,\tau}]$ is output of correction module, $\mathbf{y}_j = [y_{j,\tau-71}, \dots, y_{j,\tau}]$ ($j \in 1, 2, \dots, B$) is ground-truth at input $\mathbf{X}_{j,\tau}$ and n_c is the number of training data belonging to class c ($c \in 0, 1$).

4.2.2 Prediction Module

After training the correction module, the parameters of the trained correction module are frozen and the prediction module is trained. Figure 8 shows an overview of the training of the prediction module. The training data for the prediction module corresponds to the all binary occurrence time series.

Figure 9 shows the architecture of the prediction module. The intermediate output of the correction module is combined with the original binary occurrence time series and used for training. Since the intermediate output of the correction module is considered to contain features related to missing entries, it is used in the prediction module to make predictions considering missing entries. The prediction module outputs whether the target activity will occur within one hour after the 72-hour period used as the input of the model.

The loss function $\mathcal{L}_{\mathcal{P}}$ of the prediction module is expressed by the following equation.

$$\mathcal{L}_{\mathcal{P}} = \frac{1}{B} \sum_{j=1}^B (p_{j,\tau+1} - y_{j,\tau+1})^2$$

where B is the batch size, $p_{j,\tau+1}$ is output of prediction module and $y_{j,\tau+1}$ ($j \in 1, 2, \dots, B$) at $\mathbf{X}_{j,\tau}$ is the ground truth.

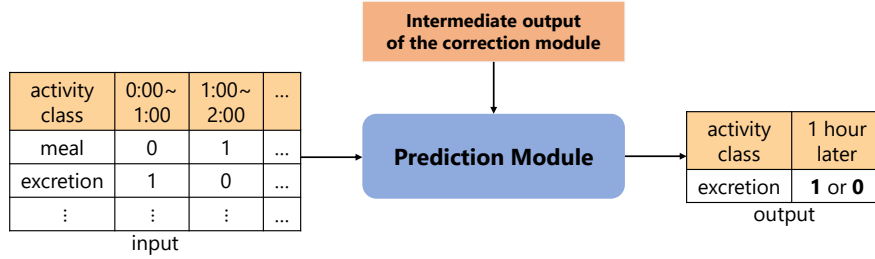


Figure 8: Overview of the training of the prediction module

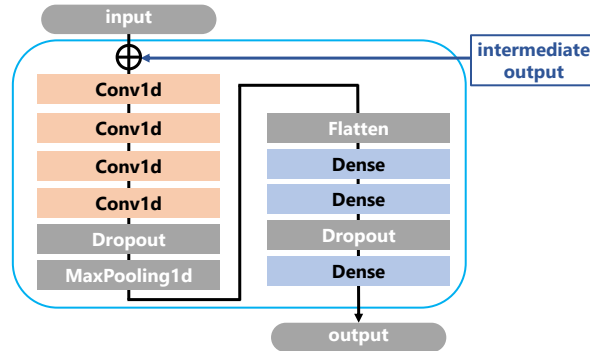


Figure 9: The architecture of the prediction module

4.3 Method of Missing Entry Creation

To create missing data for training the correction module, we utilize the binary occurrence time series with a low probability of missing entries.

For activities targeted for prediction, we create care records with missing entries by randomly invalidating occurrences with a certain probability.

Specifically, we create records with missing entries so that the distribution of the number of occurrences of the activity for the created records is equal to the distribution of the number of occurrences of the activity with possible missing entries in the original records. Since it is very difficult to simulate the actual situations of missing entries because it is necessary to consider various factors such as the busyness of caregivers, time of day, and the tendency of each caregiver to make mistakes. Therefore, we simply assume that the frequency of missing entries simply follows a binomial distribution. We use the BFGS optimization method [17] to determine the parameters of the binomial distribution that minimize the distance between the two distributions. The distance between the two distributions is defined as the inverse of the cosine similarity between the histograms of the distributions. As a result, the parameter of the binomial distribution $p = 0.2674$ minimized the distance between the distributions. Therefore, in this paper, we use $p = 0.2674$ to generate care records with missing entries.

5 Experiments

5.1 Dataset

The dataset used in the experiment was collected in an actual nursing care facility in Fukuoka, Japan. The collected data was converted into the binary occurrence time series as described in Section 3. We divide the binary occurrence time series into 72-hour time windows, which are used as inputs for the prediction model. Table 2 shows the number of data points (72-hour time windows) in the entire dataset used in the experiments and the number of data points were divided into two sets, one with a high probability of missing entries and the other with a low probability of missing entries.

Table 2: The number of data points in the dataset

	entire dataset	data points with a high probability of missing entries	data points with a low probability of missing entries
the number of data points	48178	29418	18760

5.2 Evaluation procedures

This section describes the evaluation methodology for the correction module and prediction module of the proposed method. Target activities are six activities: “Meal and medicine,” “Defecation,” “Urination,” “Nighttime user care,” “Daytime user care” and “Bathing.”

5.2.1 Correction Module

For experiments of the correction module, we use binary occurrence time series with a low probability of missing entries. As described in Section 4.3, we generated care records with missing entries by randomly altering binary occurrence time series with a certain probability. 80% of the data was used for training and 20% for testing.

The following two types of evaluation metrics were used.

- Correction success rate: Percentage of missing entries that are properly corrected
- Correction error rate: Ratio of the number of mistakenly corrected entries to the total number of corrections made

5.2.2 Prediction Module

We used data of five care recipients with relatively frequent “Defecation” and “urinary” activities in the dataset and tested it by leave-one-out cross-validation.

The following method was used for comparison. Only-Predict: It only uses the prediction module and does not use intermediate outputs of the correction module.

Because, the data used in the experiment includes missing entries, it is impossible to determine whether an entry belonging to class 0 is actually not performing caregiving activity or it is mistakenly recorded as class 0, making it impossible to calculate the precision precisely. Therefore, the recall for class 1 was used as the evaluation metric.

5.3 Results

5.3.1 Correction Module

Figure 10 shows an example of the output result of the correction module. Some missing entries are corrected appropriately while error corrections are occurring.

Table 3 shows the correction success rate and the correction error rate for each activity. For all activities, the correction success rate is approximately 0.9, indicating that the correction module is successful in correcting the created missing entries. In contrast, the correction error rate is high for “defection” and “bathing” activities. This is because the number of occurrences of these activities is fewer than that of the other activities, and therefore it is difficult to determine whether it is a missing entry or no activity has occurred.

5.3.2 Prediction Module

Figure 11 shows an example of the output result of the prediction module. Some activity occurrences are predicted properly (probabilities higher than 0.5).

Table 4 shows the recall for each activity and the average recall weighted by the number of data points of class 1 in each activity. The proposed

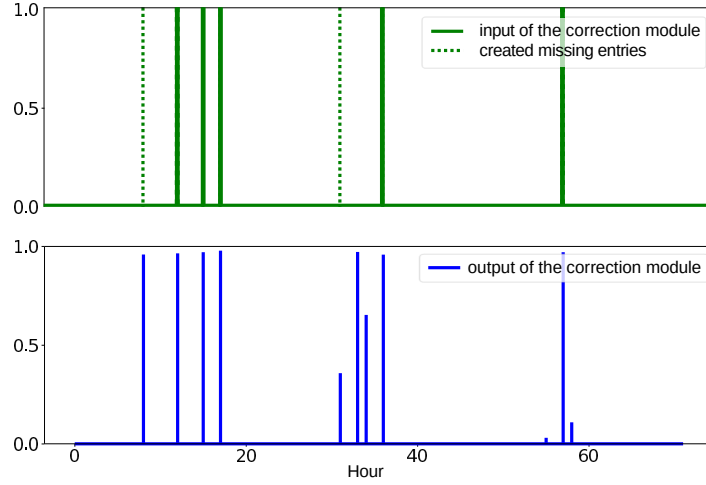


Figure 10: An example of the output result of the correction module (“Meal and Medicine” activity). The vertical axis indicates the probability of occurrence

Table 3: Experiment results of the correction module

Activity	Correction success rate	Correction error rate
Meal and medicine	0.94	0.27
Defection	0.93	0.72
Urination	0.88	0.17
Nighttime user care	0.94	0.17
Daytime user care	0.96	0.41
Bathing	0.93	0.91
Overall	0.93	0.41

method has higher recall than the comparison method for “Defecation” and “Urination” activities. Because these activities are nonperiodic and it is difficult for caregivers to schedule them in advance, the proposed method can be effective in reducing the burden on caregivers and assisting in the creation of care records. In contrast, for the “bathing” activity, the recall decreased by using the correction module. This is because the correction error rate for the “bathing” activity is particularly high, and the error correction has a negative impact on the prediction.

We used 72-hour time windows as inputs in order to consider the periodicity of the activities on the consecutive days. Using longer time windows may increase the prediction accuracy. However, using longer time windows decrease the amount of training data, making it difficult to train the neural network.

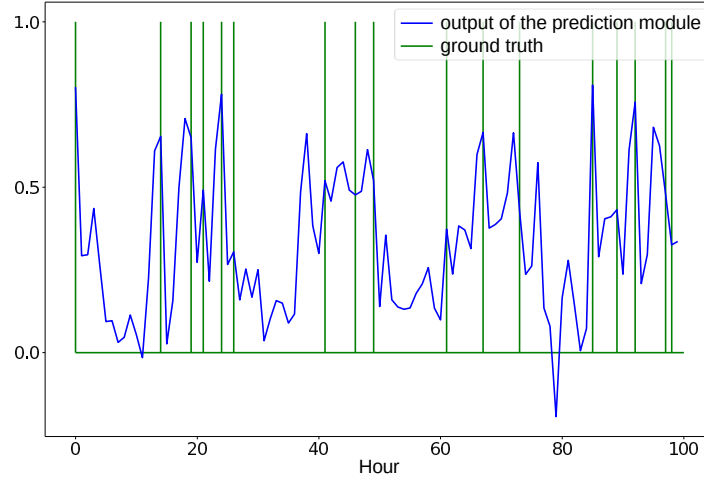


Figure 11: An example of the output result of the prediction module (“Meal and medicine” activity). The vertical axis indicates the probability of occurrence

Table 4: Recall for each activity

Activity	Proposed Method	Only-Predict
Meal and medicine	0.65	0.68
Defection	0.27	0.18
Urination	0.32	0.25
Nighttime user care	0.67	0.66
Daytime user care	0.86	0.88
Bathing	0.34	0.54
Weighted average	0.52	0.50

6 Conclusion

With the recent introduction of ICT in nursing care facilities, prediction of nursing care activity using electronic nursing care records has been studied. Besides, missing entries are unavoidable in nursing care records, and it is difficult to train a prediction model using records that contain missing entries. Focusing on missing entries in nursing care records, we proposed the correction module that corrects missing entries based on generated records with missing entries. By including the correction module in the prediction method and using the intermediate output of the correction module for prediction, we designed a caregiving activity prediction method that is robust against missing entries. We evaluated the effectiveness of the proposed method using data obtained from actual nursing care facilities. As a result, we confirmed that the proposed method is effective for “Defecation” and “Urination” activities. Because these ac-

tivities are nonperiodic and it is difficult for caregivers to schedule them in advance, the proposed method can be effective in reducing the burden on caregivers.

In the future, we plan to consider improving the prediction model to more effectively capture trends in missing entries by reconsidering the intermediate outputs of the correction module used in the prediction module.

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